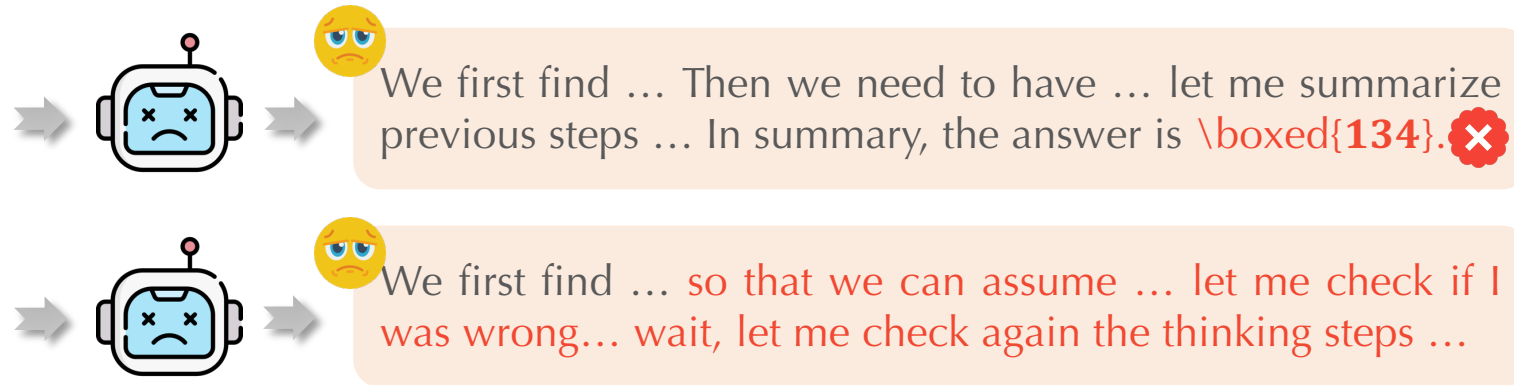


### (a) Unreliable LLM

? There exist real numbers  $x, y > 1$ , such that  $\log_x(y^x) = \log_y(x^{4y}) = 10$ . Find  $xy$ .  
👉 Solvable Problem

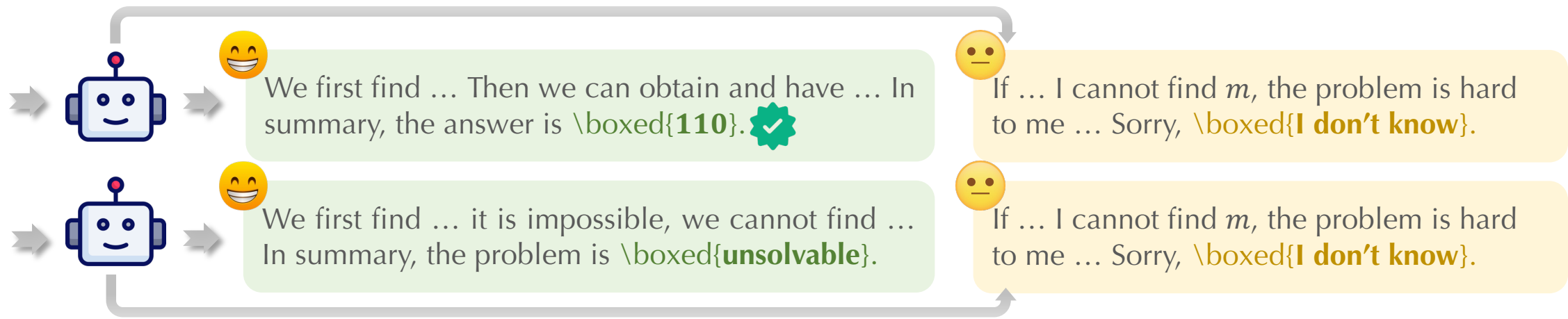
? There exist real numbers  $x, y > 1$ , such that  $\log_x(y^x) = 10$ . Find  $xy$ .  
👉 Unsolvable Problem



### (b) Reliable LLM

? There exist real numbers  $x, y > 1$ , such that  $\log_x(y^x) = \log_y(x^{4y}) = 10$ . Find  $xy$ .  
👉 Solvable Problem

? There exist real numbers  $x, y > 1$ , such that  $\log_x(y^x) = 10$ . Find  $xy$ .  
👉 Unsolvable Problem



### (c) Response Preference

